

Milestone Part 2

Abstract

- Domain = SRKW
 - Pod
 - Matrilineal
 - Observed association relationships
 - SRKW relationships include pod membership, matrilineal kinship, and social associations.
- Problem
 - Individual and family records are different from sighting/encounter records. Both are needed to query familial and observed relationships together.
- Review
 - Whales
 - Relationships
 - Encounters
 - Time
 - Location
 - Evidence
 - Examine how existing sources represent these records and compare their strengths and limitations.
- Model
 - Kinship
 - Pod
 - Recorded co-occurrence
 - Develop a draft ER model integrating relationship and sighting information while distinguishing kinship from recorded co-occurrence.
- Purpose
 - Explore observed associations and potential association candidates through shared associations and kinship.
- Repository
 - GitHub

Southern Resident killer whales (SRKWs) have multiple types of relationships, including pod membership, matrilineal kinship, and social associations. Genealogical and relationship information must be considered together with sighting and encounter information to comprehensively query familial and social relationships among individuals. This paper reviews how existing sources represent these records and compares their strengths and limitations. Based on this review, OrcaLink proposes a draft data model that integrates relationship and sighting information to explore the observed associations and potential association candidates based on shared associations and kinship. The project repository is available on GitHub.

Intro

- Identification issue
 - Report says whole pod/matriline present
 - Dataset may assign to individual ID
 - Need to find difference from presence from confirmed individuals
- Association
 - Same sighting does not mean a close social bond
 - evidence/source attached
 - Relationship can be interpreted
- Need to include
 - How whale identification and encounter recording developed
 - Different ways the source organize the information
 - What each representation captures
 - What needs to get simplified
 - Explain that in ER model

Southern Resident killer whales (SRKWs) generally travel in matrilineal family groups but also form social associations [4]. Encounter and photo-identification records have been collected since 1976 [1]. Individual records describe whales' identities, demographic information, and matrilineal relationships [6], while sighting records identify individuals observed together, along with dates and locations [7]. However, some individual identifiers in the sighting data are assigned based on reports of an entire pod or matriline rather than individually confirmed observations.

Family membership and observed associations represent different types of relationships. Although previous studies have investigated demographic and social association data [4], examining individual records alone makes it difficult to jointly query kinship, repeated co-sightings, and indirect connections across multiple encounters. Recorded co-sightings also does not necessarily establish a close social relationship. OrcaLink aims to integrate these relationships into a data model that supports queries about both direct and indirect associations among individuals while retaining their sources and identification information.

This paper reviews the history of individual identification and encounter recording and compares existing representations of whale identities, family relationships, and sightings. The review examines what information these representations capture and what is missing or simplified. OrcaLink's ER model will be based on these findings.

History

Photo-identification [<https://www.whaleresearch.com/orca-id>]

- Michael Bigg's work in 1970s
- Identifying individuals using dorsal-fin shape and saddle-patch markings
- Recognize same whales across encounters
- Recording technology changed from film to digital

CWR's first encounter

[<https://www.whaleresearch.com/post/2016/04/08/our-very-first-encounter-april-8-1976>]

- April 8, 1976
- Encounter 1
 - K and L pod near Dungeness Spit
 - Handwritten logbook
 - Film photographs
- First J pod encounter
 - <https://www.whaleresearch.com/history-old>
 - April 16, 1976
- Firsts encounters in records
 - Does not mean first sightings

Whale Museum: [<https://whalemuseum.org/pages/the-whale-hotline-sightings-network>]

- Spring 1976 started
- People phoned in to report
- Toll-free number introduced in 1977
- Current process
 - Reports online, phone and email
 - Staff work to confirm sightings
 - Confirmed sightings entered in database

Earlier record in our dataset: [The Whale Museum dataset provided for this project]

- Earliest dated row
 - Dec 6, 1948
 - "Pod = Orcas"
 - "Source = TWM-SA-Pub"
 - No time
 - No individual ID
- Predates hotline network
- Does not explain original recording method
- "Orcas" alone does not confirm SRKW identifications
- Modeling implication
 - Keep the source and level of identification with each record

Individual photo-identification developed in the 1970s through Michael Bigg's work, using photographs of dorsal fins and saddle patches to distinguish whales [<https://www.whaleresearch.com/orca-id>]. CWR's first documented encounter was on April 8, 1976, when K and L pods were observed near Dungeness Spit. The encounter was recorded in a handwritten logbook and through film photographs [<https://www.whaleresearch.com/post/2016/04/08/our-very-first-encounter-april-8-1976>]. Its first J-pod encounter followed on April 16, 1976 [<https://www.whaleresearch.com/history-old>]. These are the first encounters documented by CWR, rather than the first-ever sightings of these whales. Early film photographs were developed and reviewed to identify individuals, while current identification uses digital photographs examined on computers [<https://www.whaleresearch.com/orca-id>].

The Whale Hotline public sighting network started in spring 1976, allowing members of the public to report orca sightings in Washington State by telephone. A toll-free number was later introduced in 1977. Today, reports can be submitted online, by phone, or by email. The Whale Museum gathers these reports, and staff work to confirm sightings before entering them into the database [<https://whalemuseum.org/pages/the-whale-hotline-sightings-network>]. The dataset provided for this project also contains earlier records, with the earliest dated row being December 6, 1948. However, the row does not explain how the original observation was recorded [The Whale Museum dataset].

Recording methods have therefore developed from handwritten logs, film photographs, and telephone reports to digital photographs and database records. For OrcaLink, combining these records requires keeping their sources and distinguishing reports of whale groups from records identifying individual whales. This allows family and sighting information to be queried together while keeping the basis of each recorded connection clear.

Comparative Analysis

- Scope
 - Compare
 - CWR's public identification and encounter records
 - Dryad dataset
 - Whale Museum datasets
 - Compare all available representations
 - CWR encounter data requested still pending
- Main Purpose
 - CWR
 - Identify and track individual whales
 - Describe family structure and document encounters
 - Dryad
 - Organize observations for J-pod movement analysis
 - Whale Museum
 - Maintain historical sightings reports
 - Record reported whale presence across dates and locations
- Identification
 - CWR
 - Individual ID supported by photographic identification
 - Identification records connect whales to family groups
 - Dryad
 - Whale ID + sighting ID
 - Some whale IDs assigned based on whole pod or matriline reports
 - Some observations have no individual ID
 - Whale Museum
 - Our dataset records reported group rather than individual ID
 - Labels
 - Pod names
 - Uncertain group
 - General label (like "Orcas")
- Relationships
 - CWR
 - Family records describe maternal relationships and group membership
 - Encounter records provide a separate type of info
 - Dryad
 - Shared sighting ID connect whale records with observations
 - Can support recorded co-sighting queries
 - Lifespan file does not provide mother/offspring relationship
 - Whale museum
 - Group reports support queries about reported presence
 - Can't directly link which whales were together
- Time and locations
 - CWR

- Public encounter have date + location
 - Need the requested data to check fully
- Dryad
 - Timestamps and coordinates
 - Time and coordinates were shifted
- Whale Museum
 - Dates and geographic fields
 - Some records don't have sighting times
- Sources and identification info
 - Keep where each record came from
 - Separate individual sighting from IDs assigned using group reports
 - Check what observation flags and source codes mean
 - Shared sighting does not mean socially close
- Implications
 - Use individual IDs to connect identity, family, and encounter records
 - Represent encounters separately from whales
 - Keep kinship separate from recorded co-sightings
 - Allow group reports without assigning every group member as individually confirmed
 - Keep source info and missing values
 - Explain which queries the data can support and which will require additional data

Source	Main Purpose	Key entities or records	Key relationships	Strengths	Limitations
CWR public records	Identify whales, describe family structure, and document encounters	Whales, pods, matriline, encounters	Family relationships, pod membership, whales recorded in encounters	Helps connect individual identity and family information across records	The complete requested encounter dataset is not yet available
Dryad dataset	Organize observations for J-pod movement analysis	Whale IDs, sighting IDs, timestamps, coordinates, lifespan records	Whale records connected through shared sighting IDs	Provides structured records for examining recorded co-sightings	Some IDs are assigned based on group reports or are missing. Timestamp and coordinates are shifted
Whale Museum dataset	Maintain historical sighting records	Reports, reported groups, dates, locations, sources	Reported groups connected to sightings	Supports analysis of reported group presence over time and across locations	The provided records do not identify individual whales

CWR's public records help describe individual whales and their family groups. Photo-identification allows the same whale to be recognized across encounters, while identification guides organize whales by pods and matriline [\[https://www.whaleresearch.com/orca-id\]](https://www.whaleresearch.com/orca-id). These records are useful for connecting identity and family information in OrcaLink. However, family membership alone does not tell us whether two whales were observed together in a particular encounter. Therefore, family relationships and recorded co-sightings need to be kept separate. Our comparison is currently limited to CWR's public records because the additional encounter data requested are still pending.

The Dryad dataset links whale records to observations through sighting IDs. Records with the same sighting ID represent whales counted together in one observation, which can support recorded co-sighting queries. However, some whale IDs were assigned based on reports of a whole pod or matriline rather than individual identification. Some observations also have missing IDs, so the presence of an ID cannot always be treated as an individually confirmed sighting. The lifespan file provides demographic information but does not include mother-offspring

relationships. The timestamps and coordinates were also shifted, making matching with other sources more difficult [<https://datadryad.org/dataset/doi:10.5061/dryad.gf1vhhn2f>].

The Whale Museum dataset focuses on historical sighting reports. The museum gathers reports and works to confirm sightings before entering them into its database [<https://whalemuseum.org/pages/the-whale-hotline-sightings-network>]. The record provided for our project describes reported groups rather than individual whale IDs. They can help answer questions about when and where a group was reported, but cannot directly show which individual whales were together. The group labels also vary in detail. A named pod provides more identification information than a general label such as “Orcas” [Whale Museum Dataset].

Each source provides a different part of the information needed for OrcaLink. CWR helps describe identity and family structure, Dryad connects whale records to observations, and the Whale Museum dataset describes reported group presence. Combining these records requires keeping their source and identification information. A report of a whole pod should not automatically be treated as confirmation that every member was individually identified. Similarly, being recorded together does not necessarily mean that two whales have a close social relationship. These distinctions will help OrcaLink query family relationships and recorded associations while making the evidence behind each connection clear.